1.

Let $\tilde{T}_i$ be a topology containing $\mathcal{B}$ on X.

Then $\mathcal{T} := \bigcap_{i=1} \tilde{T}_i$.

(T1) By definition, each $\tilde{T}_i$ is a topology, thus $\phi \in \tilde{T}_i$ and $X \in \tilde{T}_i \ \forall i$. Thus, $\phi \in \bigcap_i \tilde{T}_i$ and $X \in \bigcap_i \tilde{T}_i$.

$\Rightarrow \phi \in \mathcal{T}$ and $X \in \mathcal{T} \Rightarrow$ T1 satisfied.

(T2) Assum $\{U_i\}_{i \in I} \in \mathcal{T}$. Then $\{U_i\}$ are in the intersection of all topologies $\tilde{T}_i$. If some sets are contained in another, their union is also contained in the set. Thus, $\bigcup_{i \in I} U_i \in \bigcap \tilde{T}_i \Leftrightarrow \bigcup_{i \in I} U_i \in \mathcal{T}$. This holds for arbitrary unions of U_i.

$\Rightarrow$ T2 satisfied.

(T3) By definition, $\forall U_i \in \tilde{\mathcal{T}}$, $U_i \in \bigcap \tilde{T}_i$

$\Rightarrow$ the intersection of finite U_i is contained in $\bigcap \tilde{T}_i \Rightarrow \hat{\bigcap} U_i \in \tilde{\mathcal{T}}$.

$\Rightarrow$ T3 satisfied.

$\Rightarrow \tilde{\mathcal{T}}$ is a topology. ////

(b) $\tilde{\mathcal{T}}_{\mathcal{B}}$ defined as all U s.t. $\forall x \in U \exists B \in \mathcal{B}$ s.t. $x \in B \subset U$.

<u>Lemma 2.3</u> $U \in \tilde{\mathcal{T}}_{\mathcal{B}}$ iff $U = \bigcup_{i \in I} B_i$ for $B_i \in \mathcal{B}$

$\Rightarrow$ every $U \in \tilde{\mathcal{T}}_{\mathcal{B}}$ either equals $B_i \in \mathcal{B}$ or the union of some $B_i \in \mathcal{B}$.

Since $\tilde{\mathcal{T}}$ contains $\mathcal{B}$ and all unions of elements in $\mathcal{B}$ by (T2), $\tilde{\mathcal{T}}_{\mathcal{B}} \subset \tilde{\mathcal{T}}$.

$\tilde{\mathcal{T}}_{\mathcal{B}}$ contains $\mathcal{B}$ thus is itself one of the $\tilde{T}_i$.

Since an intersection of sets is contained in each set being intersected,

$$\tilde{\mathcal{T}} = \tilde{T}_1 \cap \tilde{T}_2 \cap \tilde{\mathcal{T}}_{\mathcal{B}} \cap \cdots \cap \tilde{T}_n \Rightarrow \tilde{\mathcal{T}} \subset \tilde{\mathcal{T}}_{\mathcal{B}}$$

$$\Rightarrow \tilde{\mathcal{T}} = \tilde{\mathcal{T}}_{\mathcal{B}}$$

∎

2.

(a)

$$B_d(x, r_n) = \{y \in X \mid d(x,y) < r_n\}$$

Let $x \in X$. Since $d \geq 0$, if we show $\forall n \in \mathbb{N}\ r_n > 0$, then $\exists\ B_d(x, r_n) \ni x$.

$$r_{n+1} = \frac{r_n}{1+\sqrt{1-r_n}} = \frac{(1-\sqrt{1-r_n})r_n}{r_n} = 1-\sqrt{1-r_n}$$

Need to show $0 < r_n < 1\ \forall n \in \mathbb{N}^+$, otherwise $\sqrt{1-r_n} \notin \mathbb{R}$.

$r_1 = \frac{1}{2} \Rightarrow r_2 = 1-\sqrt{\frac{1}{2}}$, and $1-\sqrt{\frac{1}{2}} < \frac{1}{2} \Leftrightarrow r_2 < r_1$

so $r_2 < 1$. Now, since $1-\sqrt{1-x_1} < 1-\sqrt{1-x_2}$ if $x_1 < x_2$,

$r_{n+1} < r_n\ \forall n$. Also, $\sqrt{1-x} > 0$ for $(x<1) \in \mathbb{R}^+$, thus

$0 < r_{n+1} < r_n < \cdots < r_1 < 1 \Rightarrow 0 < r_n < 1\ \forall n \in \mathbb{N}$.

and r_n is strictly decreasing $\Rightarrow r_n \to 0$ as $n \to \infty$.

$\Rightarrow \exists\ B_d(x, r_n)$ centered at $x \Rightarrow x \in B_d(x, r_n)$

$\forall x \in X \Rightarrow$ (B1) satisfied.

Let $y \in B_1 \cap B_2$ i.e. $y \in B_d(x_1, r_{n_1}) \cap B_d(x_2, r_{n_2})$

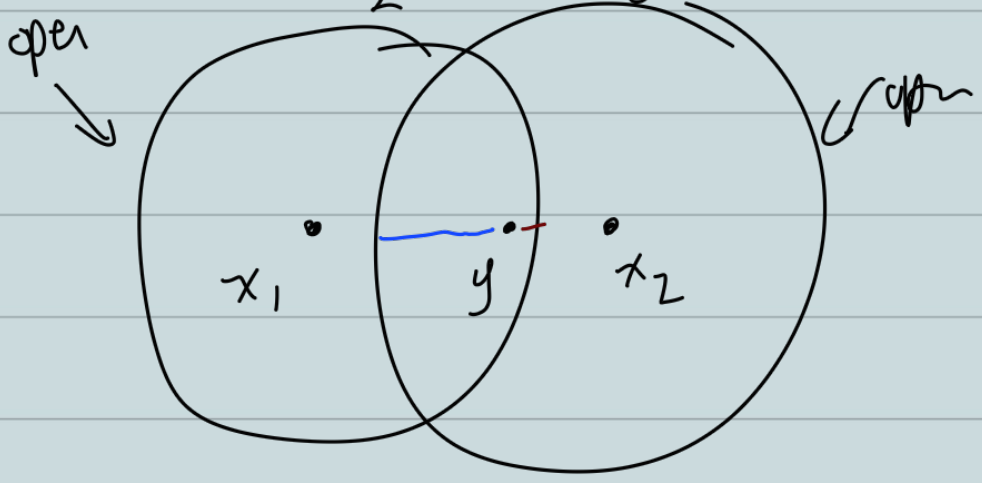


since $y \in B_d(x_1, r_{n_1})$, $d(x_1, y) < r_{n_1}$

similarly $y \in B_d(x_2, r_{n_2})$, $d(x_2, y) < r_{n_2}$

$\Rightarrow$ $r_{n_1} - d(x_1, y) > 0$

$r_{n_2} - d(x_2, y) > 0$



take the minimum, otherwise radius of new ball might leave intersection (like blue line)

(★) $\varepsilon = \min\{r_{n_1} - d(x_1, y), r_{n_2} - d(x_2, y) > 0\}$

since $\varepsilon > 0$ and $r_n \to 0$ as $n \to \infty$, we can always find $r_{n_3} < \varepsilon$.

So choose $B_d(y, r_{n_3})$. Now check if it is a subset of intersection.

Let $z \in B_d(y, r_{n_3})$. Then $d(y,z) < r_{n_3} < \varepsilon$

Using triangle property of metric,

$$d(x_1, z) \leq d(x_1, y) + d(y,z) < d(x_1, y) + \varepsilon$$

Then using (★), $\varepsilon \leq r_{n_1} - d(x_1, y)$, so

$$d(x_1, z) \leq d(x_1, y) + r_{n_1} - d(x_1, y) = r_{n_1}$$

$$\Rightarrow d(x_1, z) < r_{n_1} \Rightarrow z \in B_d(x_1, r_{n_1})$$

by anabgy, $d(x_2, z) < r_{n_2}$

$$\Rightarrow z \in B_d(x_2, r_{n_2})$$

$$\Rightarrow B_d(y, r_{n_3}) \subset B_d(x_1, r_{n_1}) \cap B_d(x_2, r_{n_2})$$

$\Rightarrow$ (B2) satisfied.

$\Rightarrow \mathcal{B}$ is basis for topology on X.

(b) Let $\mathcal{T}_{\mathcal{B}}$ be topology generated by $\mathcal{B}$ and $\mathcal{T}$ be metric topology.

Every basis element in $\mathcal{B}$ is an open ball, so every element of $\mathcal{B}$ is open in $\mathcal{T}$.

By Lemma 2.3, $U \in \mathcal{T}_{\mathcal{B}}$ iff $U = \bigcup_{i \in I} B_i$ for $B_i \in \mathcal{B}$.

By T2 of $\mathcal{T}$, it is closed under arbitrary unions, so every element in $\mathcal{T}_{\mathcal{B}}$ is in $\mathcal{T}$.

Thus $\mathcal{T}_{\mathcal{B}} \subseteq \mathcal{T}$.

Let $U \in \mathcal{T}$, $x \in U$. Then
$\exists \varepsilon > 0 \quad B(x, \varepsilon) \subseteq U$

Since $r_n \to 0$, $\exists_{n \in \mathbb{N}} r_n < \varepsilon$

↙ element $\in \mathcal{B}$

$\Rightarrow B_d(x, r_n) \subseteq B_d(x, \varepsilon) \subseteq U$

so $\forall x \in U$, $\exists B_i \in \mathcal{B}$ s.t. $x \in B \subseteq U$

$\Rightarrow U = \bigcup_{x \in U} B_d(x, r_{n(x)})$ for $r_{n(x)}$ in sequence

$\Rightarrow U \in \mathcal{T}_B$

$\Rightarrow \mathcal{T} \subseteq \mathcal{T}_B$

Thus, $\mathcal{T} = \mathcal{T}_B$ ∎

3.

(1) let $x \in X$.

$$x \in f^{-1}\left(\bigcup_{i \in I} A_i\right) \iff f(x) \in \bigcup_{i \in I} A_i$$

$$\iff \exists i \in I \text{ s.t. } f(x) \in A_i$$

$$\iff \exists i \in I \text{ s.t. } x \in f^{-1}(A_i)$$

which is exactly definition of union

$$\iff x \in \bigcup_{i \in I} f^{-1}(A_i)$$

$$x \in f^{-1}\left(\bigcap_{i \in I} A_i\right) \iff f(x) \in \bigcap_{i \in I} A_i$$

$$\iff \forall i \in I \; f(x) \in A_i$$

$$\iff \forall i \in I, \; x \in f^{-1}(A_i)$$

$$\iff x \in \bigcap_{i \in I} f^{-1}(A_i)$$

(2) let $x \in X$

$$x \in f^{-1}(Y \setminus A) \iff f(x) \in Y \setminus A$$
$$\iff f(x) \notin A \iff x \notin f^{-1}(A)$$
$$\iff x \in X \setminus f^{-1}(A)$$

(3) let $x \in X$.

$$x \in (g \circ f)^{-1}(B) \iff g(f(x)) \in B$$
$$\iff f(x) \in g^{-1}(B)$$
$$\iff x \in f^{-1}(g^{-1}(B))$$